

When Is It Worth Training Your Own Model?

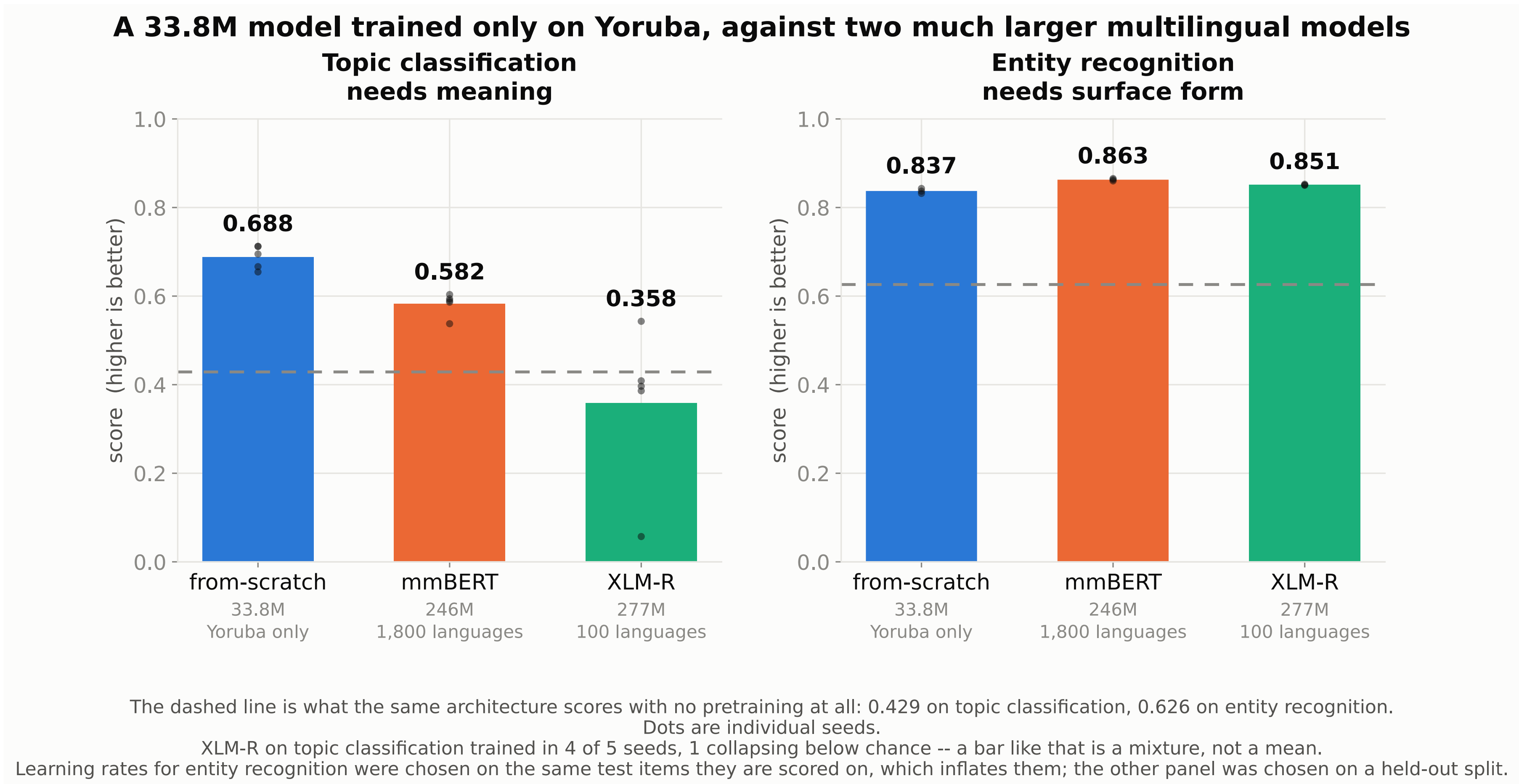
Yoruba from scratch against multilingual transfer

A2-NLP Project Team · CSED 504 · University of Washington · Summer 2026

TAKEAWAY

A 33.8M-parameter Yoruba model is ahead of mmBERT on topic classification; the strongest evidence points to vocabulary fit—not text scarcity—as the leverage.

THE RESULT



READ THE RESULT AS A PAIR

0.688

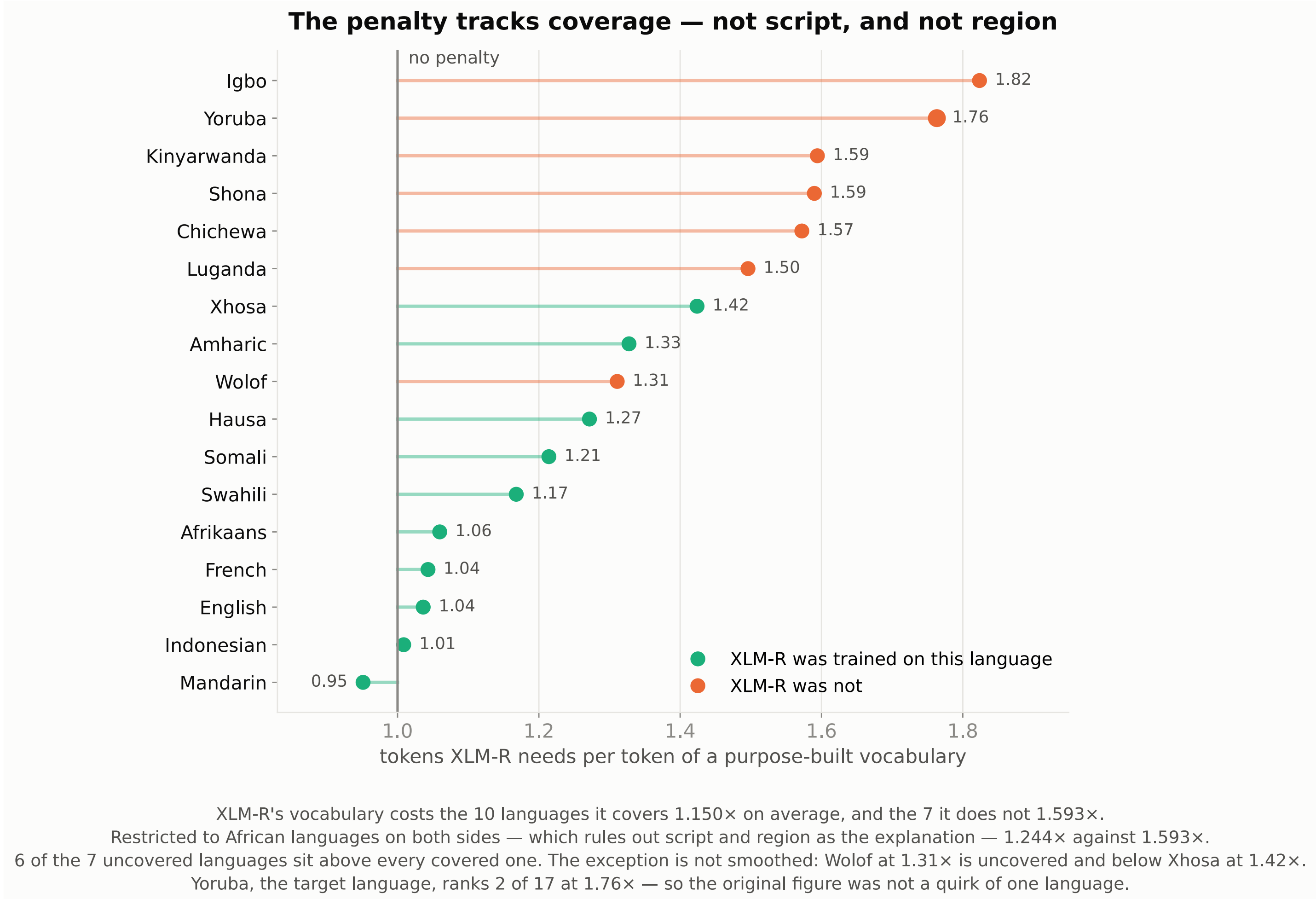
topic macro-F1
from scratch

0.837

entity F1
from scratch

Ahead of mmBERT on topic; slightly behind on entities. The disagreement is part of the finding.

THE MECHANISM



Why vocabulary fit

XLM-R needs 1.76 tokens for each token used by a Yoruba-specific 16k BPE—second-highest among 17 languages. A 128-token window holds about 44 Yoruba words with XLM-R versus 77 with the fitted vocabulary. Across languages, this penalty tracks XLM-R coverage.

Holding architecture, Yoruba text and compute fixed, swapping only the vocabulary changed downstream scores. The 16k vocabulary led the 250k XLM-R vocabulary by +0.144 on SIB-200 and +0.061 on MasakhaNER; every seed won in both tasks (four seeds per arm, exact p=0.029).

Text is not the bottleneck

FineWeb-2 contains 69.1M Yoruba tokens. At fixed compute, an English control gained nothing measurable from 64M to 1,024M tokens, and Yoruba's 16M→64M change also sat inside seed variation. At these budgets, lack of additional text does not explain the result.

Means and variability

The pretraining mean did not separate after six registered seeds (0.930 vs 0.989 bits/char; p=0.374). Variability did: sd 0.037 versus 0.145 (F=15.1, p=0.0098). A poorly fitted vocabulary is not a guaranteed loss penalty; it makes an individual run less predictable.

Scope

Causal downstream evidence is still Yoruba-only. SIB-200 has just 204 test items, and NER did not receive the same dev-selection protocol. The next decisive test is the matched-compute vocabulary swap across four or five languages spanning the coverage gradient.